

MLOps Project for a Mask R-CNN on GCP using uWSGI Flask

Components:

- GCS Bucket
- GCP CloudBuild
- Artifact Registry
- GCP Pub/Sub
- GCP CloudRun
- Google Kubernetes Engine(GKE)

Note:

The Cost Breakdown as of November 4, 2025. The costs might get updated by GCP Cloud as per their latest pricing change. The pricing may vary depending on the GCP region where the services are created. We recommend visiting the official GCP pricing page to check the current cost of the services to come up with updated guesstimates.

GCP Services Cost:

- <https://cloud.google.com/pricing>
- <https://cloud.google.com/products/calculator>

Cost Breakdown:

Components	Description	Usage	Estimated Cost
Storage	Google Cloud Storage (GCS) to store the model file	Example: 10 GB total (standard storage, multi-region)	~\$0.26 / month (\$0.026 × 10 GB)
Docker Image Storage	Artifact Registry to store Docker images for deployment	3–5 images (~500 MB each)	~\$0.10 / month
Containerization	Cloud Build used to build and package Docker images from a GitHub repo	~100 build minutes per month (default machine type)	\$0.00 / month (if exceeded → \$0.60 / 100 min)
Messaging & Event Handling	GCP Pub/Sub to handle message exchange between services	~1000 messages/day (light test traffic).	\$0.00 / month (within free 10 GB threshold)

Serverless Compute	GCP Cloud Run to host the containerized API	1 vCPU × 10 000 requests / day. Uses ~0.05 vCPU hr / day + 256 MB RAM.	~\$2.00 / month
Cluster Management	Google Kubernetes Engine (GKE) runs containerized workloads	1-node cluster (e2-medium = 2 vCPU, 4 GB RAM @ \$0.033 / hour) + control-plane fee (\$0.10 / hour).	~\$37.00 / month (Node ≈ \$23.8 + Control Plane ≈ \$13.2)

Total Estimated Cost: If an ML application runs on GKE with a CI/CD deployment process:

Cost = (GCS \$0.008 + Artifact Registry \$0.003 + Cloudbuild \$0.00 + Pub/Sub \$0.00 + GCP CloudRun \$0.066 + GKE \$1.23)
= \$1.31/day(approximate cost)

Note:

- We recommend monitoring GCP Services and deleting those that are not in use after project execution.
- These are purely estimated costs of the services. The actual charges may vary depending on several factors, such as usage duration, service configuration, and region. Please note that these are only estimates; the actual cost will depend on how the services are utilized.